

VISUAL OPERATING LANGUAGE

VERSION 4.0

A COMPLETE DESIGN TOKEN SYSTEM AND COMPONENT GRAMMAR
FOR INTELLIGENCE-STYLE GEOSPATIAL PRODUCTS

DOCUMENT	DOSSIER SPECIFICATION — VOL CLASS
STANDARD	RFC 7946 (GeoJSON) COMPLIANT
PROJECTION	WGS 84 / EPSG:4326
CLASSIFICATION	UNCLASSIFIED // FOUO
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ETHOS & TABLE OF CONTENTS

Raw Substrate, Hyper-Visible Data. The interface simulates a physical, heavily-handled intelligence dossier. The base is utilitarian and drab; the data layers are explosive and precise. Every design decision descends from three governing rules:

RULE 1: PHYSICAL CONSTRAINTS

UI elements behave like physical objects bound together. Heavy cardstock, index tabs, and tension bands organize content. Z-index is physical depth.

RULE 2: THE UTILITARIAN CANVAS

Backgrounds are raw materials — kraft paper, terrain grey, industrial ink. Never pristine white, never digital dark mode.

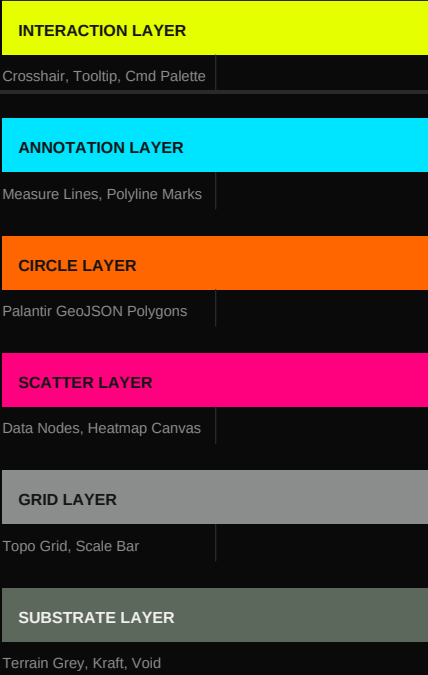
RULE 3: TACTICAL ILLUMINATION

Critical data cuts through the muted substrate with magnesium-flare intensity. Pure, high-saturation neons against drab backgrounds.

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SYSTEM ARCHITECTURE



(1)

SUBSTRATE PALETTE

The raw physical materials. Each maps to a physical material in the dossier metaphor: void is the empty desk, ink is printed text, terrain is map topography, kraft is

#0A0A0A	#151515	#1E1E1E	#2A2A2A	#333333	#5D685C
VOID --sub-void	INK --sub-ink	CHARCOAL --sub-charcoal	GRAPHITE --sub-graphite	SLATE --sub-slate	TERRAIN --sub-terrain
#8A8D8C	#A6927E	#7B5842	#D4CFC8	#F0F0F0	
GREY --sub-grey	KRAFT --sub-kraft	LEATHER --sub-leather	BONE --sub-bone	ARCHIVAL --sub-archival	

(1.1)

TACTICAL MARKERS

#0055D4	#29A5FF	#FF6600	#FF007F	#E5FF00
BLUE --mark-blue	BLUE LT --mark-blue-lt	ORANGE --mark-orange	PINK --mark-pink	YELLOW --mark-yellow
#00FF88	#FF2233	#00E5FF	#B84DFF	
GREEN --mark-green	RED --mark-red	CYAN --mark-cyan	VIOLET --mark-violet	

USAGE RULES

- Infrastructure / base data points use Industrial Ink (#151515) — black on terrain grey.
- Pathway / connection data uses Flare Orange (#FF6600) — movement and linkage.
- Organic / high-density clusters use Marker Pink (#FF007F) — biological, distributed.
- Environmental zones use Acid Yellow (#E5FF00) — boundary, categorical, highest z-index.
- Semantic states: Red = critical, Orange = warning, Green = nominal, Blue = info, Grey = inert.

(1.2)

HEATMAP RAMP

A 9-stop continuous color ramp for density visualization. Derived from the tactical marker palette, progressing from void through violet/magenta into the orange-yellow-white spectrum. Applied via screen-blend canvas overlay on the terrain substrate.



CSS CUSTOM PROPERTIES

--heat-0	#0A0A0A	Void
--heat-1	#1a0a2e	Deep
--heat-2	#3b0764	Violet
--heat-3	#7c2d8e	Magenta
--heat-4	#c2185b	Crimson
--heat-5	#FF007F	Pink
--heat-6	#FF6600	Orange
--heat-7	#E5FF00	Yellow
--heat-8	#FFFFFF	White

(6)

DATA VISUALIZATION GRAMMAR

SCATTER PLOT

Dense clusters of solid geometric circles. No strokes, no shadows, no gradients. Pure fill only. 3-5px diameter, fixed regardless of zoom. Render order: black (z-bottom), orange, pink, yellow (z-top). Visual mass indicates concentration.

CIRCLES (PALANTIR PROTOCOL)

GeoJSON-compliant radius overlays computed as N-vertex polygon approximations on the WGS84 ellipsoid. Border-only rendering with category-coded color. Per Palantir’s published method for drawing circles on maps.

HEATMAP

Canvas-rendered radial gradient accumulation with the 9-stop ramp. Screen-blend composited over terrain substrate. Respects layer filter AND temporal cutoff state. No opacity tricks — density via color progression.

SPARKLINES

Minimal bar charts for temporal activity. Computed from real feature timestamps binned per hour. Orange bars for anomaly windows above 70% of peak.

GAUGES

SVG radial progress indicators. 3px stroke, category-colored fill arc. Center-mounted percentage + label.

(2)

TYPOGRAPHY

The duality: The Shipping Label and The Coordinate. Display type is massive, tightly cropped, and commanding — a classification stamp on the face of the dossier. Data type is small, dense, and precise — coordinates etched into the margin of a terrain map.

DISPLAY — DM SANS / HELVETICA NEUE

--type-display

GEOINT

Feature Collection — Point Geometry

DATA — JETBRAINS MONO

--type-mono

```
{ "type": "FeatureCollection",
  "features": [{ "type": "Feature",
    "geometry": { "type": "Point",
      "coordinates": [-73.955, 40.735] } } ] }
```

LAT: 40.735291 | LNG: -73.955108 | ALT: 0m
TIMESTAMP: 2026-04-04T14:30:00Z

TYPE SCALE

--fs-hero	clamp(3rem, 8vw, 7rem)	Title cards, section heroes
--fs-display	clamp(1.5rem, 3vw, 2.5rem)	Section headings, stat values
--fs-heading	clamp(0.9rem, 1.5vw, 1.1rem)	Panel headers, sidebar titles
--fs-body	0.75rem (12px)	Primary content, card titles
--fs-caption	0.6rem (9.6px)	Secondary text, descriptions
--fs-micro	0.5rem (8px)	Metadata, timestamps, labels

(3)

STATUS BADGES & SEMANTIC STATES

Fixed-vocabulary status indicators. Each maps to a single semantic state. Never create custom badge colors — every status in the system resolves to one of these nine.

CRITICAL

WARNING

NOMINAL

INFO

ACTIVE

TRACK

MEASURE

ANNOTATE

INERT

SEMANTIC STATE MAPPING

●	--state-critical	#FF2233	System failure, immediate action required
●	--state-warning	#FF6600	Degraded, stale data, approaching threshold
●	--state-nominal	#00FF88	All systems operational, live feed confirmed
●	--state-active	#E5FF00	Currently selected, in focus, primary action
●	--state-info	#29A5FF	Informational, system notification
●	--state-inert	#8A8D8C	Disabled, historical, no current relevance

(4)

ALERT COMPONENTS

Border-left severity encoding. 3px left border maps to the semantic state color. Internal layout: status dot + message text.

●

SIGINT intercept detected — priority escalation require

●

Coverage gap identified in sector 7-G

●

All feeds nominal — 535 features active

●

New GeoJSON payload received from upstream

(5)

INTEL CARDS

The primary content primitive. Every intelligence item renders as a card with consistent anatomy: border-left severity encoding, type badge, timestamp, title, and coordinate metadata. In v4 every intel card is a live button — clicking routes to the related entity's dossier.

CARD ANATOMY

GEOINT

14:32:00Z

Infrastructure density anomaly — NW quadrant

40.7553, -73.9801

INT-TYPE BADGE — uppercase, 5px, grey

TITLE — Display font, bold, white

COORDINATES — Mono, blue_lt

SEVERITY BORDER — 3px, maps to priority state

PRIORITY STATES

OSINT

13:18:00Z

New pathway correlation identified

40.7401, -73.9654

DEFAULT

border-left: 3px solid

MASINT

11:45:00Z

Thermal signature shift — Zone perimeter

40.7480, -73.9520

HIGH

border-left: 3px solid

SIGINT

09:20:00Z

RF emission cluster detected — Sector 3-A

40.7350, -73.9550

CRITICAL

border-left: 3px solid

INTELLIGENCE TYPE TAXONOMY

SIGINT	Signals Intelligence — RF emissions, electronic intercepts
GEOINT	Geospatial Intelligence — satellite, aerial, map data
MASINT	Measurement & Signature Intelligence — thermal, spectral
OSINT	Open Source Intelligence — public data, web sources
HUMINT	Human Intelligence — ground reports, agent debriefs
ELINT	Electronic Intelligence — radar, emitter signatures
IMINT	Imagery Intelligence — photo, video analysis

(6.1)

SCATTER PLOT PROTOCOL

Density is Data. Do not use opacity to show heatmaps. Allow solid dots to overlap and occlude each other; the visual mass of a color indicates concentration. In v4 every node carries a timestamp; hover reveals category, coordinates, and observation time.

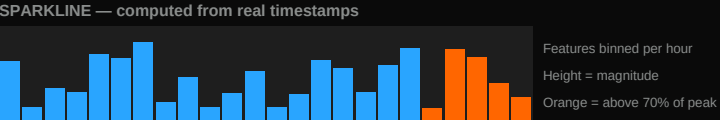
NODE ANATOMY



- Shape: Perfect circle (border-radius: 50%). No exceptions.
- Scale: 3px to 5px diameter. Fixed regardless of zoom level.
- Effects: ZERO. No drop shadows, no strokes, no gradients. Pure fill only.
- Background: Must sit on muted substrate. Never pure white or black.
- Z-order: Render infrastructure first (bottom), pathway, organic, zone last (top).
- Hover: scale(3.5) + category-colored glow. Flare tooltip shows cat + coords + time.
- Filter: CSS scale(0) transition for toggle-off. Applies to BOTH category and time filters.

(6.2)

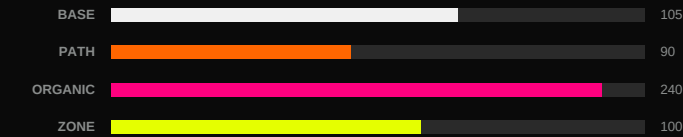
DATA VISUALIZATION PRIMITIVES



RADIAL GAUGE



DENSITY BARS



(7)

GEOJSON PROJECTION ENGINE

All geospatial data flows through an RFC 7946 compliant pipeline. The engine converts WGS 84 coordinates [longitude, latitude] to canvas positions [x%, y%] within a bounding box. The inverse function supports coordinate readout from cursor position. Per Palantir's published method, GeoJSON canvas positions are approximations of arcs on the WGS84 ellipsoid. Accounts for latitude-dependent longitude compression.

PROJECTION FUNCTIONS

```
// Forward: [lng, lat] -> [x%, y%]
function project(lng, lat) {
  x = ((lng - BBOX.minLng) / (BBOX.maxLng - BBOX.minLng)) * 100
  y = (1 - (lat - BBOX.minLat) / (BBOX.maxLat - BBOX.minLat)) * 100
}

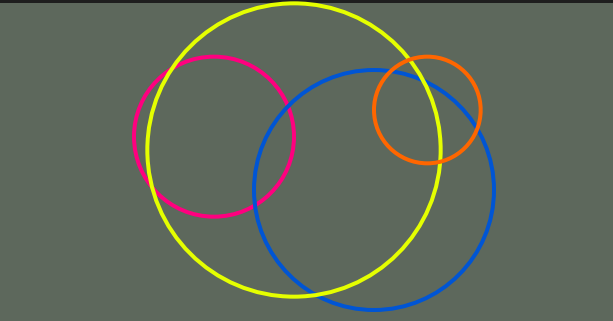
// Inverse: [x%, y%] -> [lng, lat]
function unproject(xPct, yPct) {
  lng = BBOX.minLng + (xPct / 100) * (BBOX.maxLng - BBOX.minLng)
  lat = BBOX.minLat + (1 - yPct / 100) * (BBOX.maxLat - BBOX.minLat)
}
```

RFC 7946 OUTPUT — v4 includes timestamp property

```
{
  "type": "FeatureCollection",
  "features": [{
    "type": "Feature",
    "geometry": {
      "type": "Point",
      "coordinates": [-73.955108, 40.735291]
    },
    "properties": {
      "category": "organic",
      "timestamp": "2026-04-04T14:30:00Z"
    }
  }]
}
```

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PALANTIR CIRCLE PROTOCOL



- c-pink • 0.8km
- c-blue • 1.0km
- c-yellow • 1.5km
- c-orange • 0.6km

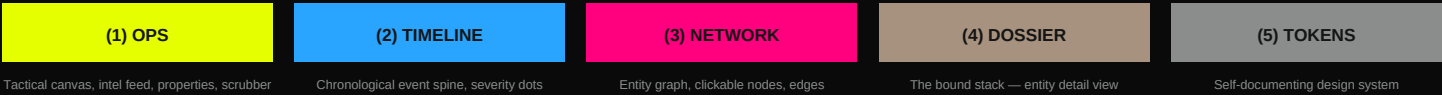
RADIUS COMPUTATION

```
latRadius = radiusKm / 111
lngRadius = radiusKm / (111 * cos(lat))
pixelR = radius / BBOX_span * 100
```

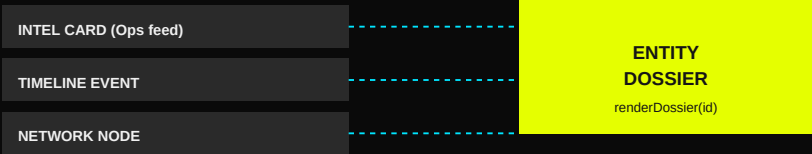
(9)

VIEW ARCHITECTURE / TEMPORAL ENGINE

v4 consolidates the system into five operational views routed through the classification strip. All views share the same entity registry, feature dataset, and temporal state — clicking an intel card, timeline event, or network node routes to that entity's dossier.



ENTITY ROUTING — three paths into the dossier

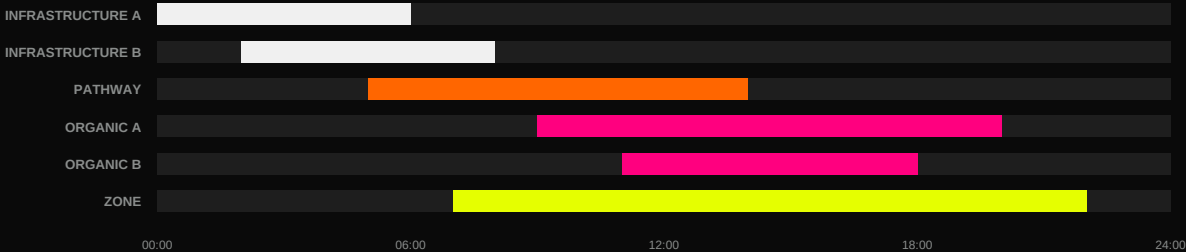


(9.1)

TEMPORAL ENGINE

Every feature carries a timestamp assigned within its cluster's birth window. The scrubber sets a cutoff hour; visibility is the conjunction of category filter AND temporal cutoff. Dragging the playhead reveals the day's intelligence accumulation in real time across scatter, heatmap, density bars, stats, and GeoJSON export.

BIRTH WINDOWS — narrative temporal structure

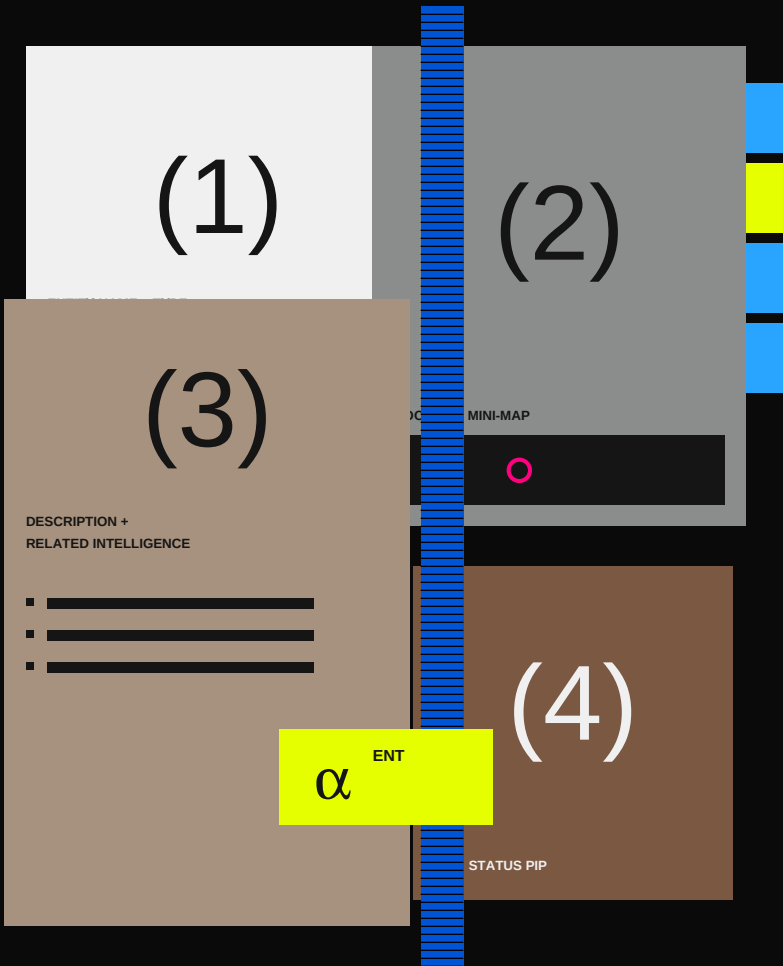


```
visible(f) = categoryFilter[f.type] && (f.ts - DAY0) / 3600000 <= cutoffHours
// conjunction of layer state and temporal cutoff – applied to all renderers
```

(10)

ENTITY DOSSIER — THE BOUND STACK

The original physical metaphor, reborn as a functional detail-view pattern. Selecting any entity renders its data live into a four-card overlapping stack bisected by the tension band. Each card carries a fixed information role; the stack is the canonical entity presentation across all DOSSIER products.



CARD (1) — ARCHIVAL

Entity designation, type, imprint metadata.

CARD (2) — TERRAIN GREY

WGS 84 coordinates, first/last observation, canvas mini-map with entity flared against dimmed dataset.

CARD (3) — KRAFT

Entity description, related intelligence events pulled live from the feed, category badge.

CARD (4) — LEATHER

Status with animated pip: live (green), stale (orange), monitor (blue).

TENSION BAND

Bisects the stack. Always on top of cards, beneath the sticker.

STICKER + TABS

Entity short-code sticker; index tabs quick-switch between entity files.

STACK RULES

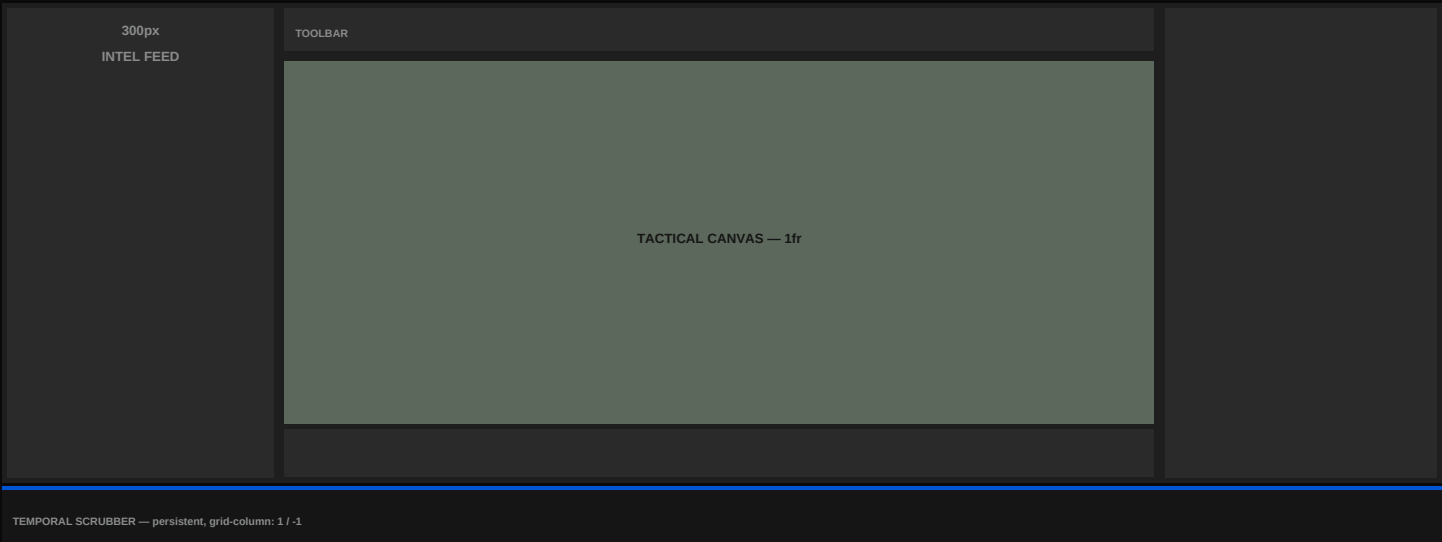
- Stack rotation: -0.8deg. Cards never perfectly aligned — desk-thrown physicality.
- Hover raises a card via elevation token (--elev-float) and z-index promotion.
- All text inside cards aligns to the rigid internal grid despite the chaotic outer composition.
- Card content is LIVE data — never static decoration. The stack is a render target.

(11)

LAYOUT & COMPOSITION RULES

Layouts should not feel like infinite digital scrolls. They should feel like a stack of differently sized papers and cardboards thrown on a desk. Even when stacked, internal text aligns to a rigid, invisible grid.

OPERATIONAL VIEW — 3-COLUMN GRID + SCRUBBER



GRID STRUCTURE

CSS Grid: grid-template-columns: 300px 1fr 240px. Sidebars are fixed-width anchors. Canvas is fluid. Scrubber spans full width in row 2.

Z-INDEX DISCIPLINE

Substrate (1) → Grid (2) → Heatmap (3) → Scatter (5) → Circles (6) → Annotations (7) → HUD (100) → Modal (500) → Toast (800) → Strip (900) → Cmd (950).

THE BOUND STACK

For entity detail views, cards stack with absolute positioning. Slight rotations (-0.5 to -1 deg). Tension band bisects vertically, beneath the sticker, above the cards.

RESPONSIVE

Below 1100px: sidebars collapse, single-column. Canvas full width. Scrubber persistent. Stack scales to 0.8.

(12)

INTERACTION VOCABULARY

The interaction model is 'Sorting the Dossier.' The user rifles through overlapping intel cards, toggles layers like pulling file tabs, scrubs the day's accumulation, and measures distances on a physical map. Every interaction has a physical analog.

POINTER (P)

Default mode. Click scatter nodes for detail panel with observation time. Crosshair tracks cursor; coord HUD shows live lat/lng.

MEASURE (M)

Two clicks define a segment. Haversine distance rendered as dashed cyan line with km label. Toast confirms.

ANNOTATE (A)

Click to place polyline vertices; double-click commits. Persistent SVG layer. Clear with X.

LAYER TOGGLE

Legend tabs as physical file tabs. Disabled: grayscale, opacity .25, pushed back into the folder.

TEMPORAL SCRUB

Drag playhead to set cutoff hour. ALL renderers respect the time window: scatter, heatmap, stats, density, GeoJSON export. Play auto-advances; R resets to 24h.

ENTITY ROUTING

Intel cards, timeline events, and network nodes all route to renderDossier(id). The dossier is the convergence point.

FLARE TOOLTIP

Node hover: scale(3.5), category glow, INK tooltip showing category + coordinates + observation time.

COMMAND PALETTE (Cmd+K)

Fuzzy-filtered commands. All tools, all five views, system actions. Enter executes first match.

DRAG-DROP IMPORT / EXPORT (E)

Drop gejson anywhere to import with validation. Export downloads visible features — respecting BOTH category and time filters.

(12.1)

MOTION & ANIMATION TOKENS

--ease-physical	cubic-bezier(0.25, 0.46, 0.45, 0.94)	Card transitions, panel slides
--ease-snap	cubic-bezier(0.34, 1.56, 0.64, 1)	Sticker slams, toast entrances
--dur-fast	0.15s	Hover states, toggle flips
--dur-med	0.3s	Panel transitions, filter changes
--dur-slow	0.6s	Circle reveals, page entrances
prefers-reduced-motion	all animations -> 0.01ms	Accessibility override — v4 requirement

(13)

SPACING, ELEVATION & Z-INDEX

SPACING SCALE

	--sp-1	4px	Micro gaps, badge padding
	--sp-2	8px	Card internal padding
	--sp-3	12px	Panel padding
	--sp-4	16px	Section spacing
	--sp-5	24px	Sidebar padding
	--sp-6	32px	Major section breaks
	--sp-7	48px	View-level padding
	--sp-8	64px	Hero spacing

ELEVATION TOKENS

--elev-card	2px 4px 12px rgba(0,0,0,0.5)	Default card, panels
--elev-float	4px 8px 24px rgba(0,0,0,0.7)	Hover-raised, stack hover
--elev-popup	0 12px 40px rgba(0,0,0,0.8)	Cmd palette, detail panel

Z-INDEX SCALE

--z-base	1	Substrate
--z-scatter	5	Data nodes
--z-circle	6	Radius overlays
--z-annotation	7	Measure / polyline layer
--z-overlay	100	HUD, minimap, stamps
--z-modal	500	Detail panel, dropzone
--z-toast	800	Notifications
--z-strip	900	Classification strip
--z-cmd	950	Command palette

END OF SPECIFICATION

This document codifies the complete visual grammar for DOSSIER v4. Every token, component, view, and interaction pattern defined here constitutes the canonical reference for intelligence-style geospatial product interfaces. The canonical implementation is dossier_v4.html; the consumable token layer is dossier-tokens.css. Implementation deviations require documented justification against these rules.